

energy_aware_controller.v

```
`timescale 1ns/1ps
module energy_aware_controller (
    input clk,
    input rst,
    input activity_in,
    output reg enable,
    output reg [1:0] state
);
// State encoding
parameter ACTIVE = 2'b00;
parameter IDLE   = 2'b01;
parameter SLEEP  = 2'b10;
reg [3:0] idle_counter;
// FSM
always @(posedge clk or posedge rst) begin
    if (rst) begin
        state <= ACTIVE;
        idle_counter <= 0;
    end else begin
        case (state)
            ACTIVE: begin
                if (activity_in == 0) begin
                    state <= IDLE;
                    idle_counter <= 0;
                end
            end
            IDLE: begin
                if (activity_in == 1) begin
                    state <= ACTIVE;
                    idle_counter <= 0;
                end else begin
                    idle_counter <= idle_counter + 1;
                    if (idle_counter >= 2) begin
                        state <= SLEEP;
                        idle_counter <= 0;
                    end
                end
            end
            SLEEP: begin
                if (activity_in == 1)
                    state <= ACTIVE;
            end
        endcase
    end
end
// Enable logic (instead of clock gating)
always @(*) begin
    case (state)
        ACTIVE: enable = 1;
        IDLE:   enable = 1;
        SLEEP:  enable = 0;
        default: enable = 1;
    endcase
end
endmodule
```